

Project report on

Intelligent Home Automation System using Hand
Gesture Recognition and Raspberry Pi for Seamless
Control

Submitted by

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Abstract

This project introduces an intelligent, contactless home automation system that leverages hand gesture recognition to control household appliances using a compact and cost-effective Raspberry Pi Zero 2 W. Traditional home automation solutions often rely on mobile apps, remotes, or voice commands, which may not be practical for everyone—especially for the elderly, people with disabilities, or in environments where voice or touch interfaces are inconvenient. Our proposed system provides an intuitive alternative by enabling users to interact with devices through simple hand gestures.

The core of the system involves capturing hand movements using a camera module or an Inertial Measurement Unit (IMU), processing these gestures using machine learning or rule-based algorithms, and executing corresponding control commands via IoT protocols like MQTT. The system is built to be lightweight, efficient, and capable of running on resource-constrained hardware such as the Raspberry Pi Zero 2 W.

The use of gesture recognition eliminates the need for physical contact, which is especially beneficial in hygiene-sensitive or hands-busy scenarios. The system has been tested for multiple gestures controlling various appliances, demonstrating reliable performance with low latency and high accuracy. It offers a flexible, scalable solution suitable for smart homes, assistive living environments, and energy-efficient automation.

Overall, this project bridges gesture-based interfaces with real-time IoT control, showcasing how embedded systems and AI can be integrated to improve human-device interaction and enhance daily living experiences.

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Chapter 01: Introduction

- 1.1. **Introduction:** This project focuses on developing an intelligent home automation system controlled through hand gestures. It replaces traditional control methods like switches and voice commands with gesture-based interaction, using a Raspberry Pi Zero 2 W. The system aims to make smart homes more accessible, contactless, and intuitive.
- 1.2. **Background:** With the rise of smart homes and IoT, controlling appliances has become more convenient, but many existing systems still rely on voice or mobile interfaces. These may not be ideal in certain situations—like noisy environments or for people with disabilities. Gesture recognition offers a natural, touch-free way to interact with smart devices, increasing accessibility and hygiene.
- 1.3. **Project Objectives:** To build a hand gesture-based home automation system, To use a Raspberry Pi Zero 2 W as the central controller for cost-effectiveness, To implement real-time gesture recognition using computer vision or sensors, To ensure seamless control of appliances with low latency.
- 1.4. **Scope:** The project was divided into phases: planning, research, component selection, hardware setup, software development, testing, and final integration. Team members shared tasks like coding, hardware interfacing, and documentation to ensure smooth progress and on-time completion.
- 1.5. **Project Management**

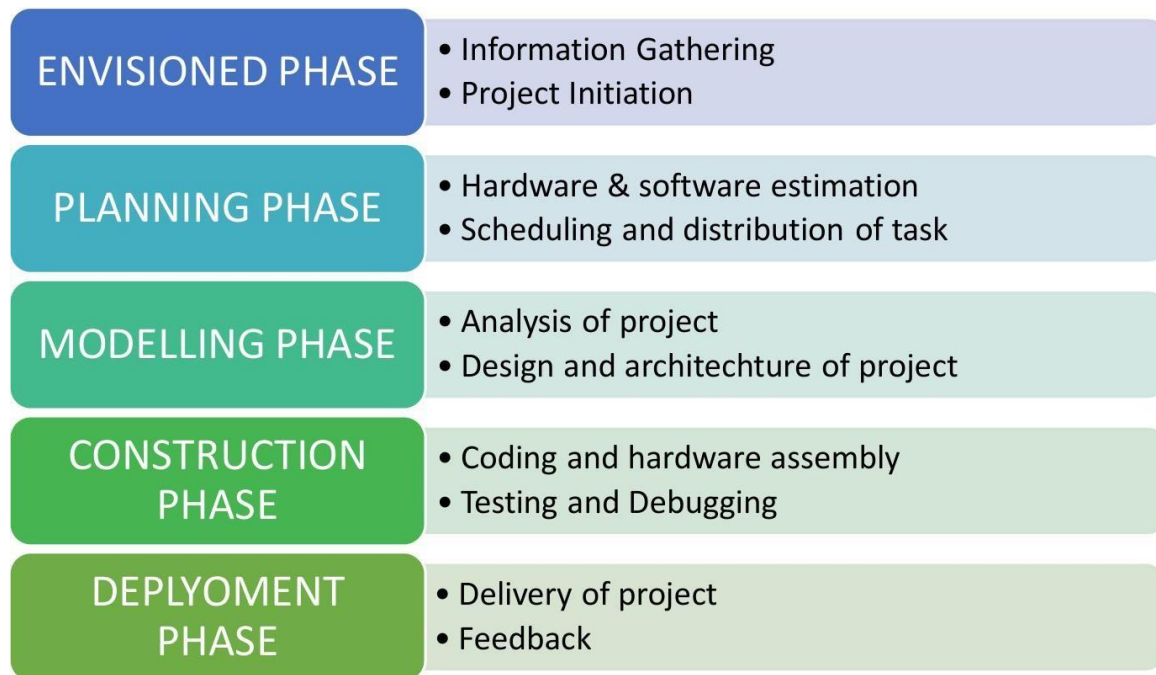


Figure 1. Model of phases in project management.

1.6. Overview and Benefits: This system enables contactless control of household appliances using intuitive gestures. It improves accessibility for elderly and disabled users and enhances hygiene by eliminating the need to touch switches. The use of affordable components makes it suitable for wide adoption in smart homes and institutions.

1.7. Organization of the Report:

The report is organised into the following chapters. Each chapter is unique on its own and is described with necessary theory to comprehend

Chapter 02: Background Review & Survey

2.

2.1. Related Works:

Several research studies and projects have explored gesture recognition and home automation individually. Gesture recognition systems often use image processing techniques, machine learning models, or motion sensors like IMUs to detect and interpret hand movements. These systems have been implemented in fields like robotics, gaming, and human-computer interaction.

On the other hand, home automation systems have advanced with the integration of IoT technologies, allowing control of appliances through smartphones, voice assistants, and centralized dashboards. Projects using platforms like Arduino, ESP modules, or full-size Raspberry Pi devices are common, but they usually depend on either voice or app-based inputs.

However, few projects combine gesture recognition directly with real-time IoT-based home automation, especially on a lightweight and cost-effective platform like the Raspberry Pi Zero 2 W. This project fills that gap by integrating gesture input with wireless device control, offering a unique solution that is affordable, contactless, and user-friendly.

Chapter 03: Theoretical Aspects

3.

3.1. Internet of Things (IoT):

The Internet of Things (IoT) refers to a network of interconnected devices that can collect and exchange data over the internet. In this project, IoT enables smart control of home appliances by linking them with a Raspberry Pi that interprets hand gestures and sends control signals.

3.2. Features of IoT:

- a) Intelligence: IoT devices process and analyze data to make intelligent decisions, like turning on a fan when a gesture is detected.
- b) Connectivity: IoT ensures devices like sensors and relays stay connected through protocols like Wi-Fi or MQTT.
- c) Dynamic Nature: Devices continuously adapt to user commands and changing environments in real-time.
- d) Enormous Scale: IoT can support a huge number of connected devices, allowing future expansion of the automation system.
- e) Sensing: Sensors (e.g., camera or IMU) detect hand gestures, which act as input signals for the system.
- f) Heterogeneity: IoT systems integrate different types of hardware and software components, like Raspberry Pi, relays, and communication modules.

- g) Security: Securing device communication and preventing unauthorized control is crucial in smart homes.

3.3. Advantages of IoT

- a) Communication: Devices can exchange information easily, enabling efficient control and monitoring.
- b) Automation and Control: IoT allows hands-free control of devices based on predefined gestures.
- c) Information: Real-time gesture data helps the system decide which appliance to activate.
- d) Monitoring: IoT systems can monitor appliance status remotely through dashboards or logs.
- e) Efficiency: Reduces energy waste by automating control and allowing optimized appliance use.

3.4. Disadvantages of IoT

- a) Compatibility: Different IoT devices may not always work together without additional configuration.
- b) Complexity: Setting up and maintaining the system requires technical knowledge.
- c) Privacy/Security: Connected devices can be vulnerable to hacking if not properly secured.
- d) Safety: System failures or incorrect gesture recognition may lead to unintended appliance control.

3.5. Application areas of IoT: Smart Homes (like this project), Healthcare (remote patient monitoring), Agriculture (smart irrigation),etc.

3.6. IOT Technologies and Protocols:

a) Bluetooth

Used for short-range device-to-device communication; low power.

b) Zigbee

A low-power wireless standard ideal for mesh networks in smart homes.

c) Z-Wave

Another low-power protocol popular in home automation; offers better interoperability.

d) Wi-Fi

Used in this project to connect Raspberry Pi with the home network for fast data exchange.

e) Cellular

Used in large-scale IoT deployments for remote monitoring, not typically used in homes.

f) NFC (Near Field Communication)

Used for secure, close-proximity data exchange—commonly seen in payments or access control.

g) LoRaWAN

Best for long-range, low-power communication in smart agriculture or city-wide IoT systems.

3.7. Project Layout

Figure 2. Layout of project module

3.7.1. Brief Description :

The project titled “**Intelligent Home Automation System Using Hand Gesture Recognition and Raspberry Pi for Seamless Control**” is designed to enable contactless control of home appliances using simple hand gestures. The system uses a Raspberry Pi Zero 2 W as the central controller, which processes input from a camera or motion sensor to recognize specific gestures.

Once a gesture is identified, the system translates it into a control signal using IoT protocols (like MQTT or Wi-Fi) to operate devices such as lights, fans, or other appliances through relays. This makes daily tasks more convenient, especially for the elderly, disabled, or in situations where touch or voice control is impractical.

The project combines gesture recognition, embedded systems, and IoT technology to create an affordable, efficient, and scalable smart home solution.

Chapter 04: Hardware Requirements

4.

4.1 Raspberry Pi Zero 2 W (Primary Controller)

The Raspberry Pi Zero 2 W is the central processing unit of the system. It handles gesture recognition, communication, and control of home appliances.

Features:

- Broadcom BCM2710A1 quad-core Cortex-A53 (ARMv8) 64-bit SoC @ 1 GHz
- 512 MB LPDDR2 SDRAM
- Built-in 2.4 GHz IEEE 802.11 b/g/n wireless LAN and Bluetooth 4.2
- Mini HDMI port, micro USB OTG port
- 40-pin GPIO header (unpopulated)
- Low power consumption and compact form factor ideal for embedded systems

Pin Configuration:

- 40 GPIO pins for interfacing sensors, relay modules, and other peripherals
- Power pins (3.3V and 5V), ground pins

- Communication pins (I2C, SPI, UART)
-

4.2 Camera Module (Input Device)

- The camera module captures hand gestures.
 - It connects to the Raspberry Pi via the CSI (Camera Serial Interface) port or USB.
 - Supports 1080p video capture, sufficient for real-time gesture recognition.
-

4.3 Sensors

Your project primarily uses vision-based gesture recognition via the camera, but optional sensors can be integrated in future upgrades.

- **4.3.1. Inertial Measurement Unit (IMU) (Optional)**
 - Sensors like MPU6050 can be used to detect hand movement in 3D space.
 - Communicates with Raspberry Pi via I2C.
- **4.3.2 Environmental Sensors (Optional)**

- Temperature or pressure sensors can be added for expanded smart home features, not currently essential.
-

4.4 Relay Module

- Used to switch home appliances ON or OFF.
 - Controlled by the Raspberry Pi GPIO pins.
 - 2 or 4-channel relay modules operating at 5V.
 - Provides electrical isolation between the Pi and high voltage devices.
-

4.5 Block Diagram of the Proposed System

- The block diagram illustrates:
User Gesture → Camera Module → Raspberry Pi Zero 2 W (Gesture Recognition) → GPIO → Relay Module → Appliance
-

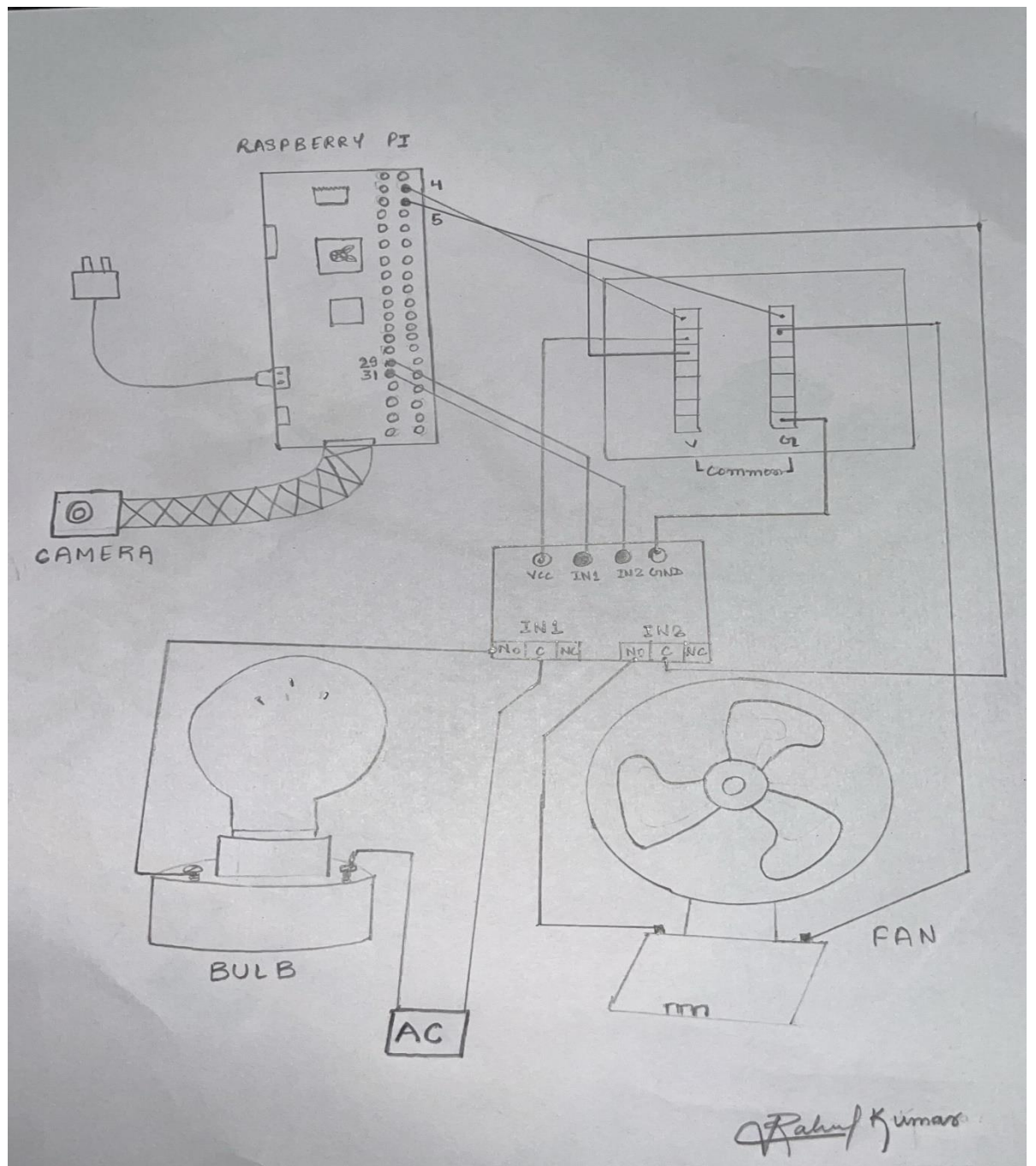
4.5.1 Working of the System

1. User performs hand gestures in front of the camera.
2. The camera sends video feed to the Raspberry Pi.

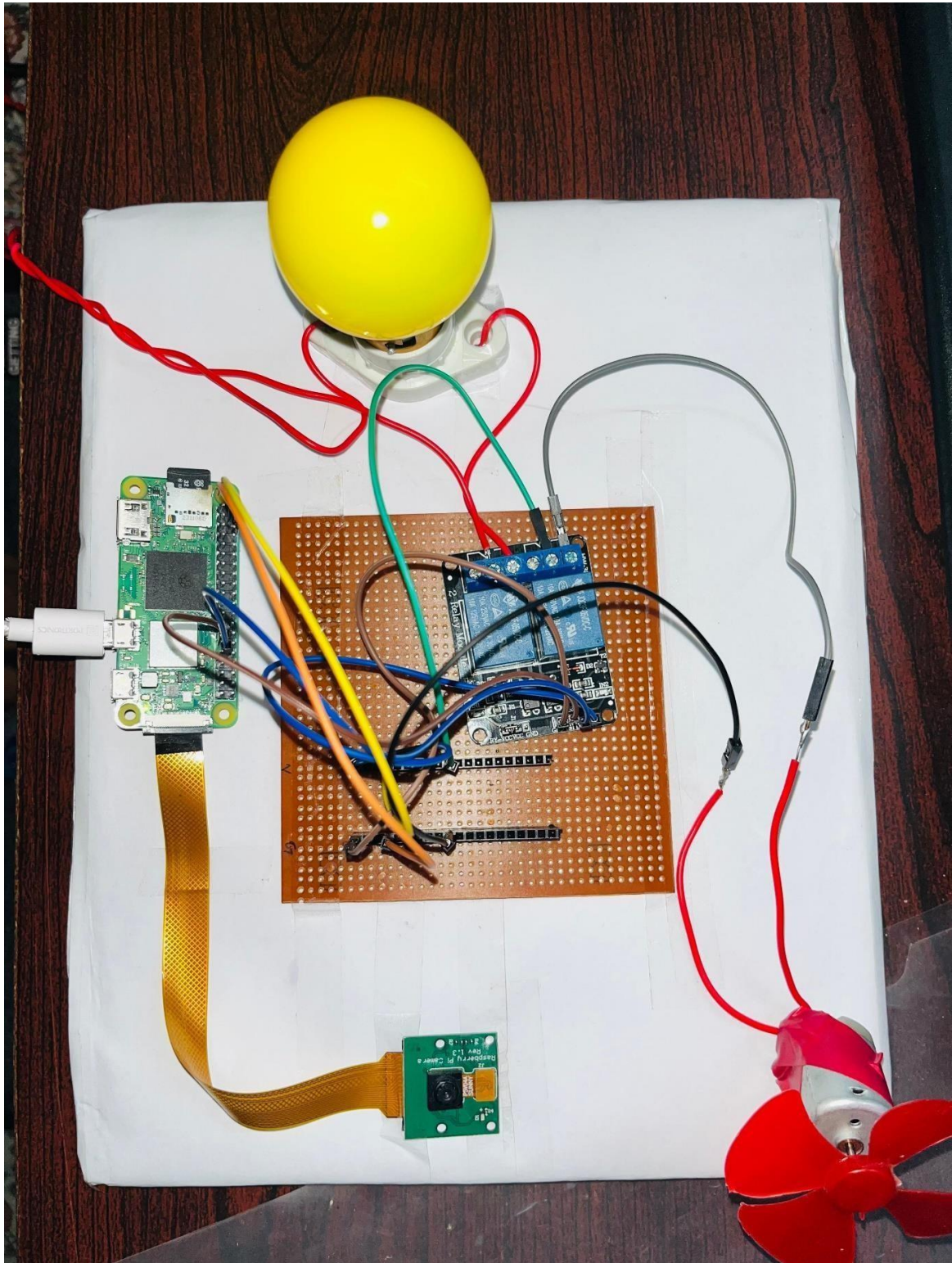
3. Raspberry Pi processes the feed using gesture recognition algorithms.
 4. Recognized gestures trigger signals sent via GPIO pins.
 5. Relay module activates corresponding appliances (lights, fans).
 6. Appliances respond instantly without physical switches.
-

4.5.2 Circuit Diagram

- Connect Raspberry Pi GPIO pins (e.g., GPIO17, GPIO27) to relay inputs.
- Connect camera to Raspberry Pi CSI or USB port.
- Relay output terminals wired to appliance's power line (with appropriate safety).
- Power Raspberry Pi and relay module with stable 5V power supplies.



WORKING PROJECT MODEL:-



4.5.3 Components Required

Sl. No.	Component and Specification	Quantity
1	Raspberry Pi Zero 2 W	1
2	Raspberry Pi Camera Module	1
3	2-Channel 5V Relay Module	1
4	Jumper Wires (Male-to-Female,Female-to-Female)	15+
5	Power Supply (5V, 2.5A for Pi, separate for relay if needed)	1-2
6	Veroboard	1
7	Home Appliances (Light bulbs, fans, etc.)	As needed

Chapter 05: Software Requirements

5.

5.1. Thonny IDE (Python):

Thonny is an easy-to-use Python Integrated Development Environment (IDE) recommended for beginners and embedded system developers alike. It comes pre-installed on Raspberry Pi OS, making it an ideal choice for developing and debugging Python scripts in this project.

For this project, Thonny IDE is used to write, test, and debug the Python programs that handle:

- Capturing input data from the camera module or sensors.
- Implementing hand gesture recognition algorithms using libraries like OpenCV and TensorFlow Lite.
- Controlling GPIO pins to operate relay modules connected to home appliances.
- Managing MQTT or other IoT communication protocols for seamless remote control.

Thonny's simple interface and debugger help streamline development and troubleshooting during the system's implementation.

5.2. Logic and Flowchart:

Logic of the System:-

1. Initialization:

- Start the Raspberry Pi and initialize camera or sensor input.

- Set up GPIO pins and establish communication with relay modules.
- Connect to the Wi-Fi network and initialize MQTT broker/client for IoT communication.

2. Data Acquisition:

- Continuously capture frames from the camera or read sensor data (IMU).

3. Gesture Recognition:

- Preprocess the captured frame (e.g., resizing, noise reduction).
- Detect and extract hand features using computer vision techniques.
- Classify the gesture using a pre-trained machine learning model or rule-based approach.

4. Decision Making:

- Map the recognized gesture to a corresponding home appliance command (e.g., turn light ON/OFF).

5. Control Output:

- Send signals to the GPIO pins to trigger relay modules.
- Activate or deactivate the connected appliance.

6. Feedback Loop:

- Monitor appliance status and provide feedback (optional).
- Continue gesture detection for next command.

7. Error Handling:

- Handle unrecognized gestures or sensor errors.

Flowchart:-

Start



Initialize Camera & GPIO



Connect to Network & MQTT Broker



Capture Gesture Input



Preprocess Input Data



Recognize Gesture?



Yes No



Map Gesture to Command Display Error / Ignore



Send Control Signal to GPIO



Activate/Deactivate Appliance



Loop Back to Capture Gesture Input

Chapter 06: Project development & Testing Aspects

6.

6.1. The development of the Intelligent Home Automation System using Hand Gesture Recognition and Raspberry Pi for Seamless Control was carried out in multiple structured phases:

7. Development Phases:

1. Component Setup:

- Raspberry Pi Zero 2 W was configured with the latest Raspberry Pi OS.
- Required libraries like OpenCV, NumPy, and GPIO libraries were installed.
- Relay modules and camera hardware were connected and tested for functionality.

2. Gesture Recognition Programming:

- A basic hand gesture recognition system was implemented using Python and OpenCV.
- Hand landmarks were tracked and mapped to predefined gestures.

- Lightweight ML models were optimized for real-time performance on Raspberry Pi Zero 2 W.

3. Control Integration:

- Each recognized gesture was linked to specific GPIO outputs.
- Relay circuits were controlled through GPIO to switch connected appliances.

4. IoT Communication (Optional):

- MQTT protocol was configured for remote monitoring and control.
- MQTT broker setup ensured seamless communication between Pi and dashboard/client.

5. User Interface (Optional Phase):

- A basic GUI or command-line display was created for debugging and feedback.

8. Testing Aspects:

1. Gesture Accuracy Testing:

- Multiple gestures were tested under different lighting conditions and hand distances.

- Accuracy was recorded and fine-tuned using threshold adjustments and image preprocessing.

2. Relay Control Testing:

- GPIO signals were tested with multimeters and verified with actual appliance switching.
- Relay activation/deactivation was synchronized with gesture inputs.

3. Latency and Performance Testing:

- Time taken from gesture detection to appliance control was measured.
- Optimizations were made to reduce system response time.

4. Error Handling:

- Incorrect gestures and unexpected inputs were gracefully ignored or flagged.
- Fallback behavior was tested to avoid accidental switching.

5. Power Consumption Monitoring:

- Total system power draw was measured to ensure compatibility with household usage.

Chapter 07: Conclusion & Future Scope

7.

- 7.1. **Result:** The intelligent home automation system using hand gesture recognition and Raspberry Pi Zero 2 W successfully demonstrates real-time control of household appliances using predefined hand gestures. The system was able to detect gestures accurately and trigger appliances through GPIO-controlled relays with minimal latency. The prototype achieved reliable performance in a compact, low-cost, and power-efficient design, making it suitable for smart home applications.
- 7.2. **Conclusion:** This project provides a touchless and user-friendly solution to home automation by replacing traditional interfaces with hand gestures. It enhances accessibility, particularly for elderly and physically challenged individuals, and offers a hygienic alternative in environments where contact should be minimized. By combining Raspberry Pi with computer vision and IoT protocols, the project proves that smart automation can be both affordable and efficient, even on resource-constrained hardware.
- 7.3. **Limitations:** Power control is limited to basic ON/OFF operations, No cloud integration or remote monitoring in the current version, environmental factors like lighting and background can affect gesture detection.
- 7.4. **Further Enhancement and Future Scope:** Improve system robustness by adapting it to function under different lighting and background conditions, Develop a user

interface (web or mobile app) to customize gesture-action mappings, Integrate cloud platforms for remote access, analytics, and scheduling, Add voice recognition as a secondary control method for hybrid interaction, Expand the system to recognize a wider range of gestures using deep learning models.

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Appendix 01

A01.1. Code Listing

7.4.1. Listing 1.1: GPIO and Device Initialization

This section initializes GPIO output devices (a bulb and a fan) connected via a relay board to pins 5 and 6 of the Raspberry Pi

```
import cv2
import numpy as np
import time
from gpiozero import OutputDevice
from picamera2 import Picamera2
import atexit

# ===== GPIO Relay Setup =====
bulb = OutputDevice(5, active_high=False, initial_value=False)
fan = OutputDevice(6, active_high=False, initial_value=False)

bulb_state = False
fan_state = False
last_toggle = time.time()
```

Listing 1.2: Graceful Shutdown Configuration

Ensures that the GPIO devices and OpenCV windows are properly closed when the program exits.

```
# ===== Graceful Shutdown =====
def cleanup():
    bulb.off()
    fan.off()
    cv2.destroyAllWindows()

atexit.register(cleanup)|
```

Listing 1.3: Camera Initialization and Configuration

Initializes the PiCamera2 with a preview configuration and waits for the camera to warm up.

```
# ===== Camera Setup =====
picam2 = Picamera2()
picam2.configure(picam2.create_preview_configuration(main={"size": (640, 480)}))
picam2.start()
time.sleep(2)
```

Listing 1.4: Gesture Detection Logic

A function to classify hand gestures based on the contour area from image processing.

```
# ===== Gesture Detection =====
def detect_gesture(contour):
    area = cv2.contourArea(contour)

    if area > 12000:
        return "palm"
    elif 4000 < area <= 9000:
        return "closed"
    else:
        return "unknown"
```

Listing 1.5: Main Loop — Frame Capture and Processing

Captures video frames, processes the region of interest (ROI), and applies image thresholding and contour detection.

```
# ===== Main Loop =====
gesture_history = []
HISTORY_LENGTH = 5
frame_count = 0
PROCESS_EVERY = 2
last_detected_gesture = "none"

while True:
    frame = picam2.capture_array()
    frame = cv2.cvtColor(frame, cv2.COLOR_RGB2BGR)
    frame = cv2.flip(frame, 1)
    roi = frame[100:400, 100:400]
```

Listing 1.6: Image Processing and Gesture Recognition

Processes every nth frame, applies Gaussian blur, thresholding, dilation, and finds contours for gesture recognition.

```
if frame_count % PROCESS_EVERY == 0:
    if roi.size == 0:
        continue

    gray = cv2.cvtColor(roi, cv2.COLOR_BGR2GRAY)
    blur = cv2.GaussianBlur(gray, (15, 15), 0)

    thresh = cv2.adaptiveThreshold(
        blur, 255, cv2.ADAPTIVE_THRESH_GAUSSIAN_C,
        cv2.THRESH_BINARY_INV, 11, 2
    )
    kernel = np.ones((3, 3), np.uint8)
    thresh = cv2.dilate(thresh, kernel, iterations=2)

    contours, _ = cv2.findContours(thresh, cv2.RETR_TREE, cv2.CHAIN_APPROX_SIMPLE)
```

Listing 1.7: Gesture Action Trigger and Relay Toggle

Analyzes the largest contour and toggles the appropriate relay if a stable gesture is detected.

```
if contours:
    contour = max(contours, key=cv2.contourArea)
    area = cv2.contourArea(contour)
    print(f"Contours found: {len(contours)} | Largest area: {area}")

    if area > 2000:
        epsilon = 0.0005 * cv2.arcLength(contour, True)
        approx = cv2.approxPolyDP(contour, epsilon, True)
        cv2.drawContours(roi, [approx], -1, (0, 255, 0), 2)

        gesture = detect_gesture(contour)
        print(f"Detected gesture: {gesture}")

        gesture_history.append(gesture)
        if len(gesture_history) > HISTORY_LENGTH:
            gesture_history.pop(0)
```

```

    if len(set(gesture_history)) == 1 and gesture_history[0] != last_detected_gesture:
        now = time.time()
        if now - last_toggle > 1:
            if gesture == "palm":
                bulb.toggle()
                bulb_state = not bulb_state
                print("Toggled Bulb")
            elif gesture == "closed":
                fan.toggle()
                fan_state = not fan_state
                print("Toggled Fan")
            last_toggle = now
            last_detected_gesture = gesture
    else:
        gesture_history.clear()
        last_detected_gesture = "none"

```

Listing 1.8: Display Overlays and User Interface

Displays gesture text, device status, and draws ROI on screen.

```

    if 'gesture' in locals():
        cv2.putText(frame, f"Gesture: {gesture}", (10, 60),
                    cv2.FONT_HERSHEY_SIMPLEX, 1.2, (0, 255, 255), 2)
        cv2.putText(frame, f"Area: {int(area)}", (10, 90),
                    cv2.FONT_HERSHEY_SIMPLEX, 0.7, (0, 255, 255), 2)

    cv2.rectangle(frame, (100, 100), (400, 400), (255, 0, 0), 2)
    status = f"Bulb: {'ON' if bulb_state else 'OFF'} | Fan: {'ON' if fan_state else 'OFF'}"
    cv2.putText(frame, status, (10, 30),
                cv2.FONT_HERSHEY_SIMPLEX, 0.9, (0, 255, 0), 2)

```

Listing 1.9: Display Loop and Program Exit

Display the final processed frame and allows the user to quit by pressing `q`.

```

    cv2.imshow("Gesture Control", frame)
    frame_count += 1

    if cv2.waitKey(1) & 0xFF == ord('q'):
        break

```

A01.2. Main Code

```
import cv2

import numpy as np

import time

from gpiozero import OutputDevice

from picamera2 import Picamera2

import atexit

# ===== GPIO Relay Setup =====

bulb = OutputDevice(5, active_high=False, initial_value=False)

fan = OutputDevice(6, active_high=False, initial_value=False)

bulb_state = False

fan_state = False

last_toggle = time.time()

# ===== Graceful Shutdown =====

def cleanup():

    bulb.off()

    fan.off()

    cv2.destroyAllWindows()

atexit.register(cleanup)
```



```
# ===== Camera Setup =====

picam2 = Picamera2()

picam2.configure(picam2.create_preview_configuration(main={"size":
(640, 480)}))

picam2.start()

time.sleep(2)

# ===== Gesture Detection =====

def detect_gesture(contour):

    area = cv2.contourArea(contour)

    if area > 12000:

        return "palm"

    elif 4000 < area <= 9000:

        return "closed"

    else:

        return "unknown"

# ===== Main Loop =====

gesture_history = []

HISTORY_LENGTH = 5

frame_count = 0

PROCESS_EVERY = 2

last_detected_gesture = "none"
```

while True:

```
    frame = picam2.capture_array()
```

```
    frame = cv2.cvtColor(frame, cv2.COLOR_RGB2BGR) # Fix color
```

```
    frame = cv2.flip(frame, 1)
```

```
    roi = frame[100:400, 100:400]
```

```
    if frame_count % PROCESS_EVERY == 0:
```

```
        if roi.size == 0:
```

```
            continue
```

```
            gray = cv2.cvtColor(roi, cv2.COLOR_BGR2GRAY)
```

```
            blur = cv2.GaussianBlur(gray, (15, 15), 0)
```

```
            thresh = cv2.adaptiveThreshold(
```

```
                blur, 255, cv2.ADAPTIVE_THRESH_GAUSSIAN_C,
```

```
                cv2.THRESH_BINARY_INV, 11, 2
```

```
            )
```

```
            kernel = np.ones((3, 3), np.uint8)
```

```
            thresh = cv2.dilate(thresh, kernel, iterations=2)
```

```
            contours, _ = cv2.findContours(thresh, cv2.RETR_TREE,
cv2.CHAIN_APPROX_SIMPLE)
```

```
            if contours:
```

```
                contour = max(contours, key=cv2.contourArea)
```

```

area = cv2.contourArea(contour)

print(f"Contours found: {len(contours)} | Largest area: {area}")

if area > 2000:

    epsilon = 0.0005 * cv2.arcLength(contour, True)

    approx = cv2.approxPolyDP(contour, epsilon, True)

    cv2.drawContours(roi, [approx], -1, (0, 255, 0), 2)

    gesture = detect_gesture(contour)

    print(f"Detected gesture: {gesture}")

    gesture_history.append(gesture)

    if len(gesture_history) > HISTORY_LENGTH:

        gesture_history.pop(0)

    if len(set(gesture_history)) == 1 and gesture_history[0] !=
last_detected_gesture:

        now = time.time()

        if now - last_toggle > 1:

            if gesture == "palm":

                bulb.toggle()

                bulb_state = not bulb_state

                print("Toggled Bulb")

            elif gesture == "closed":

                fan.toggle()

```

```

        fan_state = not fan_state

        print("Toggled Fan")

        last_toggle = now

        last_detected_gesture = gesture

    else:

        gesture_history.clear()

        last_detected_gesture = "none"

# Display detected gesture

if 'gesture' in locals():

    cv2.putText(frame, f"Gesture: {gesture}", (10, 60),

                 cv2.FONT_HERSHEY_SIMPLEX, 1.2, (0, 255, 255), 2)

    cv2.putText(frame, f"Area: {int(area)}", (10, 90),

                 cv2.FONT_HERSHEY_SIMPLEX, 0.7, (0, 255, 255), 2)

# Draw ROI and status

cv2.rectangle(frame, (100, 100), (400, 400), (255, 0, 0), 2)

status = f"Bulb: {'ON' if bulb_state else 'OFF'} | Fan: {'ON' if fan_state
else 'OFF'}"

cv2.putText(frame, status, (10, 30),

             cv2.FONT_HERSHEY_SIMPLEX, 0.9, (0, 255, 0), 2)

```

```
# Show output

cv2.imshow("Gesture Control", frame)

frame_count += 1

if cv2.waitKey(1) & 0xFF == ord('q'):

    break
```

A01 Libraries

1. cv2 (OpenCV)

Used for real-time image processing and computer vision tasks such as capturing video frames, converting color spaces, applying filters, thresholding, and detecting contours for gesture recognition.

2. numpy

Used for numerical operations and matrix manipulations, such as creating kernels and handling image arrays efficiently.

3. time

Built-in Python library used to manage delays, track time between gesture detections, and control debounce timing to avoid false toggles.

4. gpiozero

A high-level Python library for controlling GPIO pins on the Raspberry Pi. In this project, it is used to control the relay module that switches appliances like the bulb and fan.

5. picamera2

A library provided by Raspberry Pi for interfacing with the camera module using libcamera. It is used to initialize the camera and capture live video frames.

6. atexit

A standard Python library that registers cleanup functions to be executed automatically when the program exits, ensuring GPIO pins and OpenCV windows are safely closed.

Appendix 02

A02.1. Data Sheets:

1. Raspberry Pi Zero 2 W – Data Sheet

- CPU: Quad-core 64-bit ARM Cortex-A53, 1 GHz
- RAM: 512 MB LPDDR2
- GPIO: 40-pin header (3.3V logic)
- Wi-Fi: 802.11 b/g/n
- Power: 5V @ ~2.5A
- Interfaces: CSI, mini-HDMI, USB OTG

<https://www.raspberrypi.com/products/raspberry-pi-zero-2-w/>

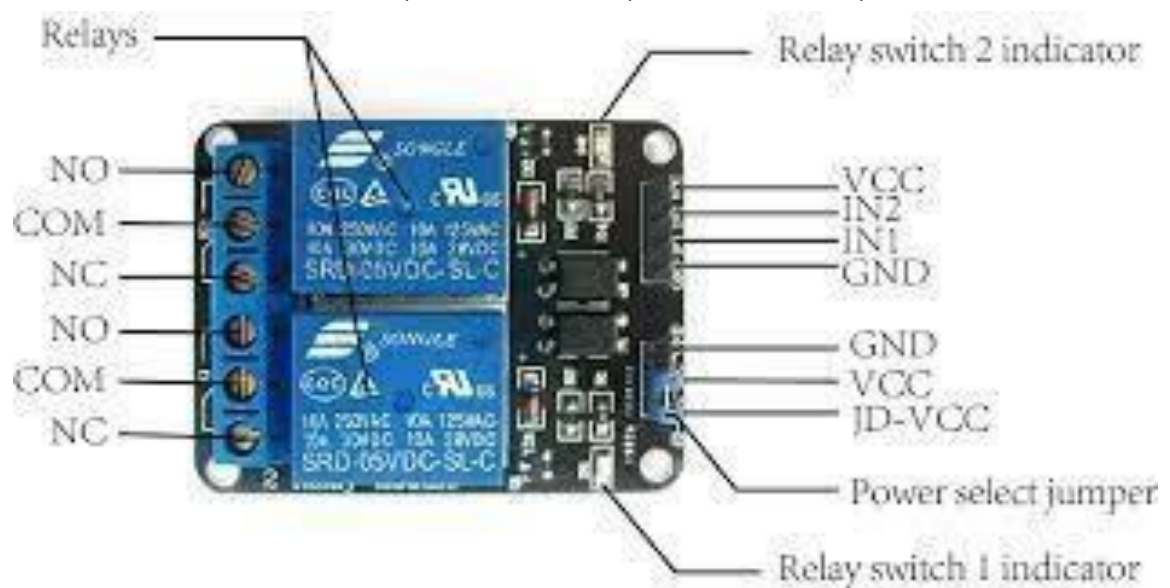
2. Camera Module (e.g., Pi Camera v2 or USB Webcam)

- Resolution: 5 MP (for Pi Camera v2)
- Interface: CSI or USB
- Power: 3.3V (CSI) / 5V (USB)
- Frame rate: Up to 1080p@30fps



3. 2-Channel Relay Module

- Operating Voltage: 5V DC
- Trigger Level: LOW (0V)
- Load Voltage: 250V AC / 30V DC
- Load Current: 10A max
- Interface: IN1, IN2, VCC, GND



4. Power Supply Adapter

- Output: 5V 2.5A (for Raspberry Pi)
- Input: 100–240V AC
- Connector: Micro USB or USB-C (depending on board)

5. Jumper Wires, Breadboard (if used)

- Standard Male-to-Female jumper wires
- Breadboard with 400 or 830 tie-points